

Supplementary Materials for: SPECTRUM: A Data-Driven, Component-Level Projection Framework for 21st-Century Sea-Level Rise

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1 Materials and Methods

Unless otherwise noted, all reported intervals are 90% Bayesian credible intervals (equal-tailed), and where 1σ is reported (e.g., the rheology correction factor), we state explicitly that it is a standard deviation.

1.1 Observational datasets

All component models are calibrated against published observational products (Table S1). We use:

- **Global mean sea level:** NASA satellite altimetry (TOPEX/Jason, 1993–present; Hamlington et al. 2020), with the Frederikse et al. (2020) tide-gauge reconstruction (1900–2018) and Dangendorf et al. (2019) reconstruction for pre-satellite validation.
- **Ocean thermal expansion:** NOAA thermosteric sea level (doi:10.7289/V53F4MVP), 0–700 m (1955–2025) and 0–2000 m (2005–2025) (*Levitus et al.*, 2012). Both records are scaled by a factor of 1.14 before fitting. *Li et al.* (2022) show that linear vertical interpolation between standard depth levels cannot reproduce the curvature of the temperature–depth profile at the vertical resolution of the historical data, and that replacing it with a multiply-rotated piecewise cubic Hermite scheme raises the 1956–2020 upper-ocean (0–700 db) thermosteric trend from 0.48 to 0.55 mm yr^{−1}, an increase of 14%. The uncorrected NOAA 0–700 m trend over 1955–2025 is 0.49 mm yr^{−1}, close to their linearly interpolated estimate, and the scaling raises it to 0.55 mm yr^{−1}, matching their value for the cubic scheme. A residual structural uncertainty of 5% of the corrected signal is retained after the scaling. *Li et al.* (2022) attribute most of the bias to the sparser vertical sampling of the pre-1990 record, so a uniform scaling reproduces their corrected trend over the full record while also raising the post-1990 rate by 14%. The 700–2000 m layer is carried by the 0–2000 m record from 2005 onward and by the deep layer of the two-layer model at earlier times. A constant 0.07 ± 0.04 mm yr^{−1} (*Purkey and Johnson*, 2010; *Desbruyères et al.*, 2016) is added over the full record for the contribution below 2000 m.
- **Glaciers:** GlaMBIE global consensus (2000–2023; GlaMBIE 2025), combining glaciological field measurements, satellite altimetry, DEM differencing, and GRACE/GRACE-FO gravimetry across all 19 RGI regions. We use 2000–2023 (23 annual points) for calibration.
- **Greenland:** Mouginot et al. (2019) SMB and discharge (1972–2018), with Mankoff et al. (2021) (1986–2023) withheld for validation. Subsurface ocean temperature from EN4 v4.2.2 with g10 bias corrections (200–500 m, Greenland peripheral waters; Good et al. 2013). Surface temperature from Berkeley Earth (*Rohde and Hausfather*, 2020).
- **Antarctica:** IMBIE satellite-era mass balance (1992–2020; Otosaka et al. 2023), partitioned by region.
- **Terrestrial water storage:** Frederikse et al. (2020) reconstruction (1900–2018).

- **Temperature scenarios:** IPCC AR6 assessed GMST projections (SSP1-2.6, SSP2-4.5, SSP3-7.0, SSP5-8.5; *Lee et al.*, 2021).

Table S1: Observational and projection datasets used in the component models. A complete dataset inventory including versions, spatial domains, and processing details is provided in `data/dataset_inventory.xlsx`.

Component	Dataset	Time range	Reference
GMSL	NASA satellite altimetry	1993–present	Hamlington et al. (2020)
GMSL	Frederikse GMSL reconstruction	1900–2018	Frederikse et al. (2020)
Thermosteric	NOAA thermosteric SL (0–700 m)	1955–present	Levitus et al. (2012)
Thermosteric	NOAA thermosteric SL (0–2000 m)	2005–present	Levitus et al. (2012)
Glaciers	GlaMBIE global consensus	2000–2023	Zemp et al. (2025)
Greenland	Mouginot SMB & discharge	1972–2018	Mouginot et al. (2019)
Greenland (val.)	Mankoff PROMICE	1986–present	Mankoff et al. (2021)
Greenland	EN4 subsurface ocean T (v4.2.2)	1970–2021	Good et al. (2013)
Temperature	Berkeley Earth GMST	1850–present	Rohde & Hausfather (2020)
WAIS	IMBIE West Antarctica	1992–2020	Otosaka et al. (2023)
EAIS	IMBIE East Antarctica	1992–2020	Otosaka et al. (2023)
Peninsula	IMBIE Antarctic Peninsula	1992–2020	Otosaka et al. (2023)
TWS	Frederikse reconstruction	1900–2018	Frederikse et al. (2020)
Projections	IPCC AR6 FACTS (med. & low conf.)	2020–2150	Garner et al. (2023)

1.2 Component models

1.2.1 Thermosteric

1.2.2 Glaciers

1.2.3 Greenland SMB

1.2.4 Greenland discharge

Surface-to-ocean transfer function.

To project T_{ocean} into the future, we calibrate a linear transfer function on annual-mean observations:

$$T_{\text{ocean}}(t) = \alpha T_{\text{surface}}^{\text{Gr}}(t) + \beta + \varepsilon(t), \quad (1)$$

where $T_{\text{surface}}^{\text{Gr}}$ is the Greenland regional surface temperature anomaly (Berkeley Earth gridded, annual mean), and ε is a zero-mean residual. Both series are anomalies relative to 1995–2005. An OLS fit over 52 common annual values gives $\alpha = 0.148 \pm 0.027$, $\beta = -0.051 \pm 0.031$ °C, $R^2 = 0.38$, and residual standard deviation $\sigma_{\varepsilon} = 0.20$ °C. The transfer function is calibrated independently of the discharge model parameters (γ, r_0, δ): the two regressions share no free parameters and are fit on different dependent variables (T_{ocean} vs. cumulative discharge).

For projections, SSP GMST trajectories are converted to Greenland regional surface temperature using an Arctic amplification factor (AA) of 3.0 ± 0.5 (1σ), consistent with

the observed ratio of Greenland-to-global warming over 1970–2021. SSP Greenland regional surface temperatures (CMIP6, rebaselined to Berkeley Earth over 1995–2005) are passed through Eq. (1) to obtain projected T_{ocean} , which then enters Eq. (??) via the delayed integral. Each of the 2,000 Monte Carlo samples draws $AA \sim \mathcal{N}(3.0, 0.5^2)$, $\alpha \sim \mathcal{N}(0.148, 0.027^2)$, and $\beta \sim \mathcal{N}(-0.051, 0.031^2)$ jointly with (γ, r_0, δ) from the discharge posterior, so that Arctic amplification, transfer-function, and discharge parameter uncertainties all propagate into the projections. A single draw of $(\alpha, \beta, \gamma, r_0, \delta)$ is applied to all SSP scenarios, so that inter-SSP differences isolate the effect of the emissions pathway from parameter uncertainty.

1.2.5 EAIS

The EAIS component uses the same level-space Bayesian framework as glaciers and the Peninsula. The rate–temperature model is

$$\dot{H}_{\text{eais}}(t) = aT(t)^2 + bT(t) + c, \quad (2)$$

where $T(t)$ is the GMST anomaly. This is fit to the IMBIE EAIS record (1992–2020, 29 annual points; *Otosaka et al.*, 2023) via emcee (32 walkers, 2000 burn-in + 4000 production steps, thinning by 2). The quadratic model yields $R^2 = 0.66$ (posterior mean: $a = 0.054$, $b = 0.035$, $c = -0.023 \text{ mm yr}^{-1}$), and the linear model yields $R^2 = 0.56$ ($b = 0.021 \text{ mm yr}^{-1} \text{ } ^\circ\text{C}^{-1}$, $c = -0.018 \text{ mm yr}^{-1}$). BIC weakly favors the quadratic ($\Delta\text{BIC} = +4.1$ ($\text{BIC}_{\text{lin}} - \text{BIC}_{\text{quad}}$, same convention as above)), but both R^2 values are too low for reliable projections given the 29-year record and large interannual variability (Fig. S1). EAIS projections instead use the literature-derived accumulation sensitivity $C_T = 60 \pm 20 \text{ Gt yr}^{-1} \text{ } ^\circ\text{C}^{-1}$ of GMST (*Frierler et al.*, 2015; *Ligtenberg et al.*, 2013), reflecting the Clausius–Clapeyron scaling of snowfall with warming ($C_T > 0$ corresponds to mass gain, i.e., negative SLR). The EAIS contribution is small (−13 to −30 mm at 2100, depending on scenario).

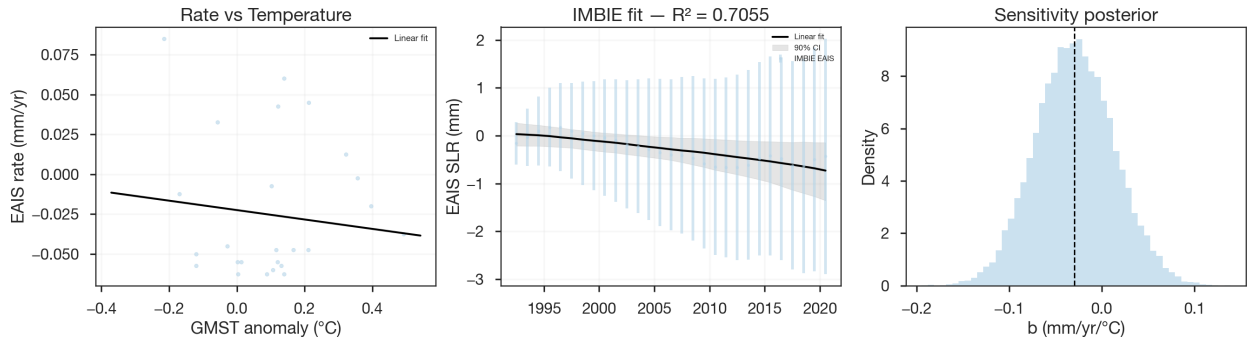


Figure S1: **EAIS model diagnostics.** (a) EAIS mass loss rate vs GMST anomaly with the fitted quadratic relationship. (b) Cumulative EAIS SLR: IMBIE observations with the model fit and 90% credible interval. (c) Posterior distribution for the temperature sensitivity b .

1.2.6 Peninsula

The Antarctic Peninsula component uses the same level-space Bayesian framework, with the rate–temperature model $\dot{H}_{\text{pen}}(t) = aT(t)^2 + bT(t) + c$ calibrated against the IMBIE

Peninsula record (1992–2020, 29 annual points; *Otosaka et al.*, 2023) via emcee (32 walkers, 2000 burn-in + 4000 production steps, thinning by 2). BIC favors the linear model ($\Delta\text{BIC} = -3.5$ ($\text{BIC}_{\text{lin}} - \text{BIC}_{\text{quad}}$)). The adopted linear model achieves $R^2 = 0.94$ with posterior median sensitivity $b = 0.055 \text{ mm yr}^{-1} \text{ }^\circ\text{C}^{-1}$ (90% CI: -0.031 – 0.138) and committed rate $c = 0.028 \text{ mm yr}^{-1}$ (90% CI: 0.003 – 0.052) (Fig. S2). As for GrIS discharge and glaciers above, this is a cumulative calibration record with correlated errors, so we report intervals from a covariance that accounts for that correlation rather than from the posterior spread, which assumes independent errors; the corrected interval for b spans zero, indicating that the temperature sensitivity is not well resolved by this record alone. The Peninsula contribution reflects combined SMB and discharge responses to warming and is small relative to other components (~ 7 – 12 mm at 2100), so this uncertainty does not materially affect the projected forecast.

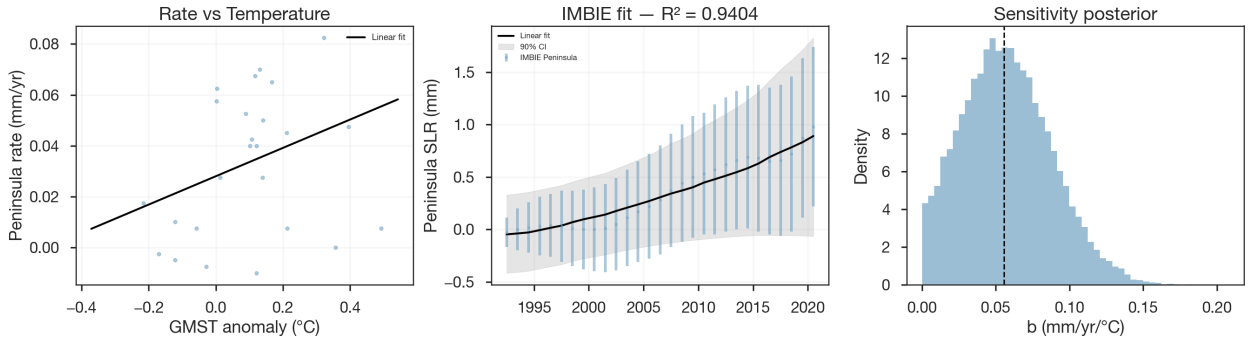


Figure S2: **Peninsula model diagnostics.** (a) Peninsula mass loss rate vs GMST anomaly with the fitted linear relationship. (b) Cumulative Peninsula SLR: IMBIE observations with the model fit and 90% credible interval; $R^2 = 0.94$. (c) Posterior distribution for the temperature sensitivity b .

1.2.7 WAIS

Ice dynamics governing WAIS mass loss involve instability mechanisms that current process models have not proven themselves capable of fully representing (*Seroussi et al.*, 2020; *Aschwanden et al.*, 2021). In the absence of reliable physical models, we take arguably the only approach available by constructing a scenario-mixture framework that represents this deep uncertainty, is grounded in the scholarly literature, and provides an accounting for recent discoveries, like the higher value of the viscous stress-exponent n (*Millstein et al.*, 2022; *Ranganathan and Minchew*, 2024) and sensitivity of sea-level rise projections to that value (*Getraer and Morlighem*, 2025; *Martin et al.*, 2026). Our framework comprises three physically motivated scenarios, each characterized by a probability weight, an endpoint distribution at 2100, a temporal trajectory, and a multiplicative rheology (n) correction. We choose to set all WAIS scenarios to be independent of the warming trajectory because there are vanishingly few data to constrain the sensitivity of WAIS to ocean thermal forcing in a time when large calving events are rapidly changing ice shelves—the floating extension of the ice sheet—and because evidence suggests ocean thermal forcing sufficient to sustain WAIS retreat is committed regardless of emissions pathway (*Naughten et al.*, 2023). We recognize

that WAIS likely retains some sensitivity to emissions pathway, but that sensitivity is likely to be small relative to the wide uncertainties arising from dynamics already underway.

Observational context

The IMBIE satellite-era record (1992–2020; *Otosaka et al.*, 2023) shows WAIS cumulative mass loss of ~ 6.6 mm SLE, with the rate increasing from ~ 0.15 mm yr $^{-1}$ in 1992 to a peak of ~ 0.44 mm yr $^{-1}$ around 2010–2012. The mass loss signal emerges from measurement noise (signal-to-noise ratio > 2) by approximately 1998. A structural break at ~ 2010 separates a period of acceleration (0.015 mm yr $^{-2}$, 1992–2010) from a period of apparent deceleration (-0.031 mm yr $^{-2}$, 2010–2020). This recent deceleration may reflect internal ocean variability in the Amundsen Sea (*Jenkins et al.*, 2018) rather than a reversal of the ice-dynamical trend. The WAIS contribution represents ~ 8 –9% of the total GMSL rate and ~ 35 –70% of the barystatic (non-thermohal) contribution.

Few if any runs in the 188-member ISMIP6 ensemble (*Seroussi et al.*, 2020) reproduce the observed trajectory of WAIS mass loss through 2020 (Fig. 2f of the main text). The ensemble 95th percentile merely tracks the lower bound of observations, and the ensemble median implies near-zero or slightly positive mass change under warming—a result that is difficult to reconcile with satellite measurements and reflects known structural biases in the underlying models (see “Known biases in ice sheet model projections” below).

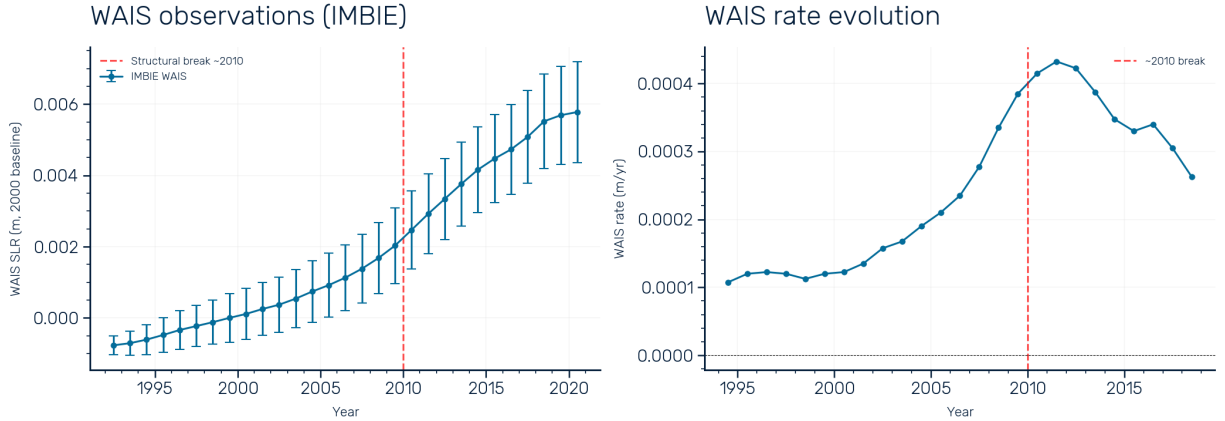


Figure S3: **WAIS observational record.** Left: cumulative WAIS sea-level contribution from IMBIE (1992–2020; *Otosaka et al.*, 2023) with measurement uncertainties, relative to the year-2000 baseline. The structural break at ~ 2010 (red dashed line) separates a period of acceleration from apparent deceleration. Right: WAIS mass loss rate evolution, showing the rate increasing from ~ 0.15 mm yr $^{-1}$ in the early 1990s to a peak of ~ 0.44 mm yr $^{-1}$ around 2010–2012, followed by a decline that may reflect internal ocean variability in the Amundsen Sea (*Jenkins et al.*, 2018).

Framework structure: two scenarios

The framework comprises two scenarios spanning the range from stable WAIS (no instability) through marine ice sheet instability (MISI), with or without the additional cascading

failure of marine ice cliffs (MICI). An earlier version of this framework separated “moderate MISI” and “MISI + MICI” into two scenarios (S2 and S3); we merge these into one, because the distinction is not physically defensible—if MISI is triggered, the amplifying processes that current ISMIP6-class models omit (calving, subglacial hydrology, the $n = 3$ vs. $n \approx 4$ rheology bias, and whatever contribution ice-cliff failure makes) all operate concurrently rather than as mutually exclusive branches:

- **S1: Status quo** ($P = 0.10$; 25–85 mm at 2100; $\alpha = 0$). No marine ice sheet instability operates. Discharge responds linearly to ocean thermal forcing, and grounding-line retreat does not trigger the positive feedback characteristic of MISI. The range is derived from a physical scaling: the lower bound extrapolates the IMBIE time-averaged rate (0.24 mm yr^{-1}) over 80 yr from the anchor year ($\sim 19 \text{ mm}$); the 25 mm lower bound rounds up to account for measurement uncertainty in the baseline rate; the upper bound applies the projected Amundsen Sea ocean warming enhancement factor ($f = [(2.15 + \Delta T_{\text{warm}})/2.15]^2 \approx 2.9$ for $+1.5^\circ\text{C}$ warming; *Naughten et al.*, 2023) to the peak observed rate, yielding a linear ramp to $\sim 85 \text{ mm}$.
- **S2: Fast WAIS (MISI + MICI)** ($P = 0.90$; 150–1300 mm at 2100; $\alpha = +3.22$). Marine ice sheet instability drives accelerating grounding-line retreat on retrograde beds, with amplification from processes that current models omit (calving, subglacial hydrology, ice damage), possibly compounded by marine ice cliff instability (MICI), in which loss of buttressing ice shelves exposes ice cliffs that exceed the yield strength of ice, triggering structural failure and rapid retreat (*Bassis and Walker*, 2012; *DeConto and Pollard*, 2016). The lower bound (150 mm) is anchored to transient-calibrated ice sheet model projections (Goldberg et al. (2026)) and observed grounding-line retreat rates (Rignot et al. (2026)), unchanged from the former MISI-only scenario. The upper bound (1300 mm) is pinned to the IPCC AR6 low-confidence Antarctic ice sheet 95th percentile under SSP5-8.5 (*Fox-Kemper et al.*, 2021): 1309 mm, rounded—so that this bound now governs the full 90% weight of the fast branch rather than only the 10%-weight MICI tail of the former three-scenario version. This anchoring is appropriate because the AR6 low-confidence projection incorporates the deep-uncertainty processes (MISI + MICI) that define this scenario; for GMSL purposes, it is immaterial whether ice loss originates from WAIS or from marine basins of EAIS—both contribute to the same global ocean—and in the AR6 low-confidence storyline, WAIS dominates but EAIS marine basins also contribute. The upper bound additionally matches the Amundsen Sea Embayment ice volume above flotation ($V_{af} \approx 1100 \text{ mm}$; *Morlighem et al.*, 2020), providing an independent cross-check estimate. The skewness ($\alpha = +3.22$) and trajectory exponent (log-mean $\ln 1.84$) are probability-weighted blends of the former MISI-only ($\alpha = +4$, log-mean $\ln 1.8$; weight 0.80/0.90 of the merged mass) and MISI+MICI ($\alpha = -3$, log-mean $\ln 2.2$; weight 0.10/0.90) branches: $\alpha = (0.80 \times 4.0 + 0.10 \times (-3.0))/0.90 = 3.22$. This composites the positive skewness motivated by the analytical and numerical results of Robel et al. (2019), who showed that the grounding-line flux nonlinearity ($q_g \propto h_g^{n+1}$, where h_g is grounding-line thickness; *Schoof*, 2007) amplifies and skews the uncertainty distribution toward greater ice loss when the nonlinearity exponent exceeds 3 (realistic values are 4–5), together with

the negative skewness reflecting skepticism that MICI operates at maximum efficiency (see below), rather than refitting to any single target percentile. The trajectory follows a power-law ramp $H(t) = H_{\text{anchor}} + H_{\text{remaining}} \cdot [(t - t_a)/(2100 - t_a)]^\beta$, where the trajectory exponent β is drawn from a log-normal distribution with median 1.84 and log-standard deviation 0.3, derived from the Schoof (2007) grounding-line flux scaling as described in the “Temporal trajectory” section below.

Scenario weights

The 10–90 weight split reflects the following assessments:

S1 = 10% (no MISI). Sustained WAIS stability requires grounding lines to stabilize despite ongoing retreat on retrograde slopes, committed ocean warming (Naughten et al., 2023), and observed acceleration. IPCC AR6 states that MISI “may be underway” in the Amundsen Sea Embayment (medium confidence; Fox-Kemper et al., 2021); Joughin et al. (2014) concluded that early-stage collapse of Thwaites is already underway; and van den Akker et al. (2025) showed that present-day ocean thermal forcing, held constant, is sufficient to deglacierate large parts of WAIS within 300–2500 yr. The 10% weight reflects the small but non-zero possibility that observed trends are reversible or dominated by internal variability.

S2 = 90% (Fast WAIS: MISI, with or without MICI). MISI is the most physically supported instability mechanism, included in IPCC medium-confidence projections (Fox-Kemper et al., 2021), and is arguably already initiated (Joughin et al., 2014; Rignot et al., 2026). Ritz et al. (2015) found MISI to be the dominant instability pathway in a probabilistic treatment of Antarctic contributions; Bamber et al. (2019) structured expert judgment produced WAIS distributions dominated by MISI-type dynamics with strongly positive skew; and DeConto et al. (2021) obtained total Antarctic median ~ 340 mm under RCP8.5 without MICI but with nonlinear acceleration above 3°C . The 90% weight reflects a consensus position that MISI is the baseline expectation for WAIS evolution, folding in the additional possibility of MICI: IPCC AR6 assigns low confidence to MICI and excludes it from medium-confidence projections (Fox-Kemper et al., 2021), and several studies have challenged the efficiency of MICI during the 21st century—Morlighem et al. (2024) found that dynamic thinning prevents cliffs from growing tall enough for sustained structural failure, requiring calving rates $\geq 25\times$ higher than physically motivated models suggest; Clerc et al. (2019) showed that realistic ice-shelf removal timescales (days or longer) raise the critical cliff height from ~ 90 m to ~ 540 m via viscous relaxation; Schlemm et al. (2022) demonstrated that mélange buttressing and embayment geometry changes are self-limiting; and a factor-of-2 error in the derivation of critical cliff height arising from an incorrect application of the deviatoric stress decomposition (cf. Eq. 1 of Bassis and Walker, 2012) calls into question the prescribed value of the critical cliff height used in projections (DeConto and Pollard, 2016). However, if our goal is to represent the scholarly literature without regard for individual views and opinions, then MICI cannot be entirely excluded; its contribution is instead folded into S2 with negative skew (α blended toward -3 by an 0.10/0.90 weight), consistent with AR6’s low-confidence assessment for that process. We note that our conclusions are qualitatively insensitive to whether MICI operates, because of its minority contribution to the blended skewness.

Statistical model for endpoint distributions

Within each scenario, the 2100 endpoint is drawn from a skew-normal distribution in log-space, $\text{SN}_{\log}(\xi, \omega, \alpha)$, where ξ and ω are solved such that the 5th and 95th percentiles match the scenario bounds. The log-space formulation is motivated by three physical considerations: (i) the dominant uncertainties (bed geometry, basal friction, ocean thermal forcing, buttressing loss) enter as multiplicative factors on grounding-line flux $q_g \propto h_g^{n+1}$ (Schoof, 2007), making a multiplicative (log-space) distribution natural; (ii) mass loss is strictly positive, which log-space enforces without truncation; and (iii) the scenarios span two orders of magnitude (25–1300 mm), and a log-space family provides scale-invariant behavior across this range. For $\alpha = 0$, the distribution reduces to a log-normal.

The skew-normal extension captures the asymmetry arising from flux nonlinearity. Robel et al. (2019) showed analytically that when the grounding-line flux exponent $n + 1 > 3$, ensemble distributions of retreating grounding lines are systematically skewed toward greater retreat: more-retreated members retreat faster (positive feedback), while less-retreated members experience weaker instability, creating a heavy right tail.

Temporal trajectory

The cumulative WAIS contribution at year t is given by

$$H(t) = H_{\text{anchor}} + (H_{2100} - H_{\text{anchor}}) \left(\frac{t - t_a}{2100 - t_a} \right)^{\beta}, \quad (3)$$

where H_{anchor} is the IMBIE observed cumulative value at the anchor year $t_a = 2020$ (the final year of the IMBIE record; $H_{\text{anchor}} = 5.8 \pm 0.7$ mm, 1σ , relative to the year-2000 baseline), H_{2100} is the scenario endpoint, and β is a trajectory exponent drawn from a log-normal distribution. For years before the anchor ($t \leq t_a$), the framework returns IMBIE observations with their measurement uncertainties rather than scenario projections. For S1 (no instability), $\beta = 1$ (constant rate, linear ramp). For S2, β has log-mean $\ln(1.84)$ and log-standard deviation 0.3, reflecting the accelerating, back-loaded character of MISI dynamics; this value is the same probability-weighted blend described above of the former MISI-only ($\ln 1.8$) and MISI+MICI ($\ln 2.2$) branches, the latter reflecting the more rapid acceleration expected when MICI supplements MISI. The power-law trajectory form is motivated by the Schoof (2007) grounding-line flux scaling $q_g \propto h_g^{n+1}$: if grounding-line retreat proceeds at roughly constant rate on a retrograde bed with slope s , then h_g decreases approximately linearly, $h_g(t) \sim h_0 - sv(t - t_0)$, and the cumulative flux integral grows faster than linearly because q_g increases as h_g decreases. For a power-law flux, dimensional analysis gives cumulative loss scaling as $(t - t_0)^{\beta}$ with $\beta > 1$, where the exact value depends on bed geometry and n ; for $n = 3$ –4, this yields $\beta \approx 1.5$ –2.5.

The trajectory exponent is corrected for rheology: the reference β is drawn assuming $n = 3$ (matching the literature from which the scaling is derived) and then adjusted to $\beta_{\text{corrected}} = \beta_{n=3} \cdot (n_{\text{draw}} + 1)/(n_{\text{ref}} + 1)$, where $n_{\text{draw}} \sim \mathcal{N}(4.1, 0.4^2)$ is drawn from the observed Glen’s law exponent distribution (Millstein et al., 2022; Ranganathan and Minchew, 2024). This correction increases β by $\sim 28\%$ at the median, producing more back-loaded (accelerating) trajectories when the ice viscosity has a higher sensitivity to stress than commonly assumed with $n = 3$.

Rheology correction

A multiplicative rheology correction is applied to all scenario endpoints. All ISMIP6 models and the literature scenarios from which our endpoint ranges are derived assume Glen’s flow law with $n = 3$, but satellite observations indicate $n = 4.1 \pm 0.4$ in fast-flowing Antarctic ice (*Millstein et al.*, 2022), supported by laboratory measurements ($n \approx 3.5$ – 3.7 ; *Fan et al.*, 2025), Greenland observations ($n \approx 4$; *Bons et al.*, 2018), and theoretical analysis of the dislocation creep regime (*Ranganathan and Minchew*, 2024). The consequence for projections has been quantified: *Martin et al.* (2026) found that using $n = 3$ when the true value is $n = 4$ underestimates 100-year sea-level rise by 21–35% in the MISIP+ and ABUMIP benchmarks, and *Getraer & Morlighem* (2025) found $32 \pm 14\%$ greater Amundsen Sea Embayment loss with $n = 4$ in an ISMIP6-framework experiment. The bias grows nonlinearly with forcing strength (*Martin et al.*, 2026).

We implement the rheology correction in Mode B (unified n -driven corrections; defined as one option in the accompanying jupyter notebook), in which a single draw of the Glen’s law exponent n per Monte Carlo sample drives both corrections through the physical link to n :

1. $n_{\text{draw}} \sim \mathcal{N}(4.1, 0.4^2)$.
2. Endpoint: $R(n) = 1 + r_0 (n_{\text{draw}} - n_{\text{ref}})$, where $r_0 = 0.28 \pm 0.07$ (1σ) is the fractional change in endpoint per unit change in n , calibrated from *Martin et al.* (2026) and *Getraer & Morlighem* (2025).
3. Trajectory: $\beta(n) = \beta_{\text{ref}} \cdot (n_{\text{draw}} + 1)/(n_{\text{ref}} + 1)$.

This couples the endpoint and trajectory corrections through n , which is more physically consistent than drawing R and the β correction independently (Mode A, defined in the accompanying jupyter notebook). At the median observed exponent $n = 4.1$, $R = 1 + 0.28 \times 1.1 = 1.31$; at $\pm 1\sigma$ in n , the correction spans from $R = 1.20$ ($n = 3.7$) to $R = 1.42$ ($n = 4.5$), consistent with the literature range of 1.21–1.35 from *Martin et al.* (2026). The correction is applied to all scenarios because the rheology bias affects every ice sheet model regardless of which instability mechanism operates. Importantly, R and β correct conceptually independent quantities: the endpoint ranges are defined at $n = 3$, and R scales the total ice lost (the endpoint magnitude), while the β correction changes when that ice is lost (the trajectory shape). These corrections are not double-counting because *Martin et al.* (2026) and *Getraer & Morlighem* (2025) report the ratio of total sea-level change at $n = 4$ to $n = 3$ —a cumulative endpoint quantity—not the trajectory shape.

Known biases in ice sheet model projections

Our scenario ranges exceed IPCC AR6 medium-confidence projections for WAIS. This is expected and physically justified by multiple documented biases in the ISMIP6 ensemble:

1. *Rheology bias.* All ISMIP6 models assume $n = 3$; observed $n \approx 4$ produces 21–35% more sea-level rise (*Martin et al.*, 2026; *Getraer and Morlighem*, 2025). Models initialized with $n = 3$ can match $n = 4$ observations in the short term by tuning a viscosity multiplier, but this compensation breaks down on projection timescales (*Martin et al.*, 2026).

2. *Missing calving.* No ISMIP6 model adequately represents iceberg calving (*Aschwanden et al.*, 2021; *Fricker et al.*, 2025), which is a first-order control on grounding-line retreat rate. The sensitivity of glacier flow to major calving events is well documented (e.g., 2–8 $\times$ speedup after Larsen B collapse; *Scambos et al.*, 2004).
3. *Initialization uncertainty.* Goldberg et al. (2026) showed that the choice of calibration data (velocity vs. surface elevation) produces a factor-of-two spread in projected Thwaites loss rates—comparable to the inter-model spread in ISMIP6. Surface-elevation calibration, which better captures inland thinning signatures of MISI, consistently produces larger mass loss than velocity calibration.
4. *Incomplete uncertainty quantification.* ISMIP6 integrates over model structure but not over parametric uncertainty within individual models, which alone could produce an interquartile range of up to 12.9 cm SLE for RCP8.5 at 2100 (*Aschwanden et al.*, 2019).
5. *Biased ensemble.* ISMIP6 models share numerical methods, parameterizations, and jointly omit key physics—they are not independent samples from the space of possible models (*Aschwanden et al.*, 2021).
6. *Historical validation failure.* Few if any ISMIP6 models reproduce the observed trajectory of WAIS mass loss; the ensemble 95th percentile merely tracks the lower bound of IMBIE observations (*Aschwanden et al.*, 2021). Under continued warming, IPCC models show slower sea-level rise rates than extrapolation of modern observations (*Hamlington et al.*, 2024).

Independent cross-check: structured expert judgment

The Bamber et al. (2019) structured expert judgment (SEJ) provides an independent, methodology-orthogonal validation target. Their WAIS-specific results under +5 °C warming give a median of 180 mm and 95th percentile of 930 mm, with strong positive skew. Our A4 mixture (median $\sim$ 410 mm, 90% CI: 60–1620 mm, after rheology correction) is wider and shifted higher than the Bamber estimates. This is expected because: (a) Bamber was published before the rheology bias was quantified (*Martin et al.*, 2026); (b) the SEJ was conducted before Morlighem et al. (2024) and recent Thwaites observations that have added to the evidence for ongoing instability; and (c) our framework explicitly applies the rheology correction, which shifts the entire distribution upward by $\sim$ 28%. The agreement in the broad distributional shape—strongly right-skewed, with median in the hundreds of millimeters—strengthens confidence in both approaches despite their different methodologies.

As a second cross-check, Rosier et al. (2025) calibrated an ice sheet model (Úa) against ASE observations using Bayesian inference with deep surrogates and projected Amundsen Sea sector contributions of 19.0 ± 2.2 mm (Paris 2 °C) and 18.9 ± 2.7 mm (RCP8.5) by 2100—statistically indistinguishable across scenarios, supporting the SSP-independence assumption adopted here. Their calibrated contribution falls within our S1 range (25–85 mm), consistent with the expectation that process models calibrated against the short observational record produce estimates at the lower end of our framework. Rosier et al. include both cliff-height- and hydrofracture-driven calving but find no evidence of unstable calving-front retreat in any

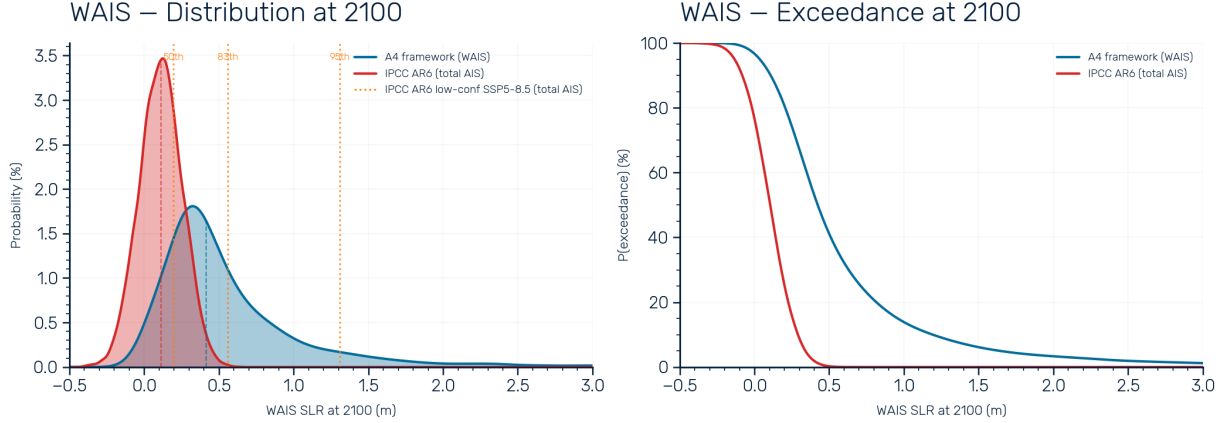


Figure S4: **WAIS endpoint distributions and exceedance probabilities: A4 vs IPCC.** Left: probability density at 2100 for the A4 WAIS framework (blue) compared with the IPCC AR6 total Antarctic projection (red). The A4 distribution is wider, with a higher median and a heavier right tail arising from the MISI skewness and rheology correction. Right: exceedance probability curves, showing that the A4 framework assigns substantially higher probability to exceeding any given threshold above ~ 0.1 m. The IPCC distribution assigns near-zero probability to outcomes exceeding 0.5 m, whereas the A4 framework assigns $\sim 40\%$ probability to this threshold.

simulation, and their creep-exponent prior ($n \in [2, 5]$, uniform) is broad but centered lower than the satellite-derived $n = 4.1 \pm 0.4$ (Millstein *et al.*, 2022). Their narrow uncertainty range (~ 5 mm, 1σ) reflects the structural constraints of a single model rather than the epistemic uncertainty that our scenario mixture model is designed to represent.

Sensitivity to scenario weights

To assess the robustness of the A4 mixture to the assigned scenario weight, we sweep $P(S1)$ —the only free weight parameter, since $P(S2) = 1 - P(S1)$ with only two scenarios—over 0–30% and examine the effect on the mixture median and 95th percentile at 2100 (Fig. S5). Over this range, the median varies from 0.45 m ($P(S1) = 0$) to 0.32 m ($P(S1) = 0.30$), a span of ~ 0.13 m around the baseline value of 0.42 m at $P(S1) = 0.10$ —small relative to the 90% credible interval (~ 1.6 m). The 95th percentile varies from 1.72 m to 1.46 m over the same range, a span of ~ 0.26 m ($\sim 16\%$ of the baseline value, 1.63 m). The insensitivity of both statistics to the scenario weight reflects the dominance of within-scenario uncertainty (controlled by the endpoint ranges and skewness) over the between-scenario weighting.

1.3 Variance decomposition

A variance decomposition using the law of total variance across the three policy-relevant SSPs (1-2.6, 2-4.5, 3-7.0) attributes 92% of total forecast variance to within-scenario uncertainty—predominantly WAIS (91% of total variance)—and 8% to between-scenario differences. Cross-component covariances are small ($+0.0014$ m², representing 0.5% of total variance), confirm-

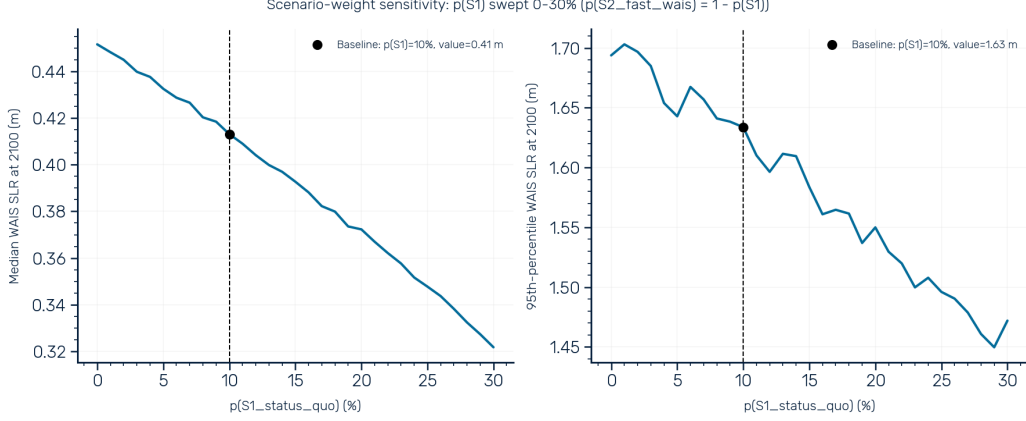


Figure S5: **Scenario weight sensitivity.** Median (left) and 95th-percentile (right) WAIS SLR at 2100 as a function of $P(S1_status_quo)$, swept from 0 to 30% ($P(S2_fast_wais) = 1 - P(S1)$). The marker indicates the baseline value ($P(S1) = 0.10$). Both statistics vary only modestly across the swept range, confirming that within-scenario uncertainty dominates over the assigned scenario weight.

ing that the independent calibration approach does not introduce material correlations.

1.4 Temperature scenarios

SSP temperature trajectories (SSP1-2.6, SSP2-4.5, SSP3-7.0, SSP5-8.5) are from IPCC AR6 assessed warming projections. Historical forcing uses Berkeley Earth monthly GMST (*Rohde and Hausfather, 2020*). All anomalies are relative to a 1995–2005 baseline (≈ 2000 reference year). Forecasts extend from 2020 to 2150, with pre-2020 values from the observational record. SSP GMST trajectories are converted to Greenland regional surface temperature using an Arctic amplification factor of 3.0 ± 0.5 (1σ), consistent with the observed ratio of Greenland-to-global warming over 1970–2021.

1.5 Prior specifications

Table S2 lists the prior distributions used in each Bayesian component model. All priors are weakly informative and are chosen to regularize the posterior without dominating the likelihood. All values in the table are reported in mm (the code operates in meters internally).

Table S2: Prior distributions for all component models. Exp = Exponential, HN = Half-Normal, HC = Half-Cauchy, LN = Log-Normal, N = Normal, SN_{log} = Skew-Normal in log-space. Parameters a , b , c in the level-space models correspond to the quadratic, linear, and constant coefficients of the rate-temperature relationship, respectively.

Component	Parameter	Prior	Source / justification
<i>Thermosteric (two-layer ODE, 6 free parameters)</i>			
	$\log \tau_u$	$N(\log 8, 0.5^2)$	LN median 8 yr; Geoffroy et al. range
	$a, b_u, b_d, c, \sigma_{\text{extra}}$	flat within physically wide bounds	
<i>Glaciers, EAIS, Peninsula (level-space Bayesian, 5 free parameters each)</i>			
	a ($d\alpha/dT$)	$\text{Exp}(\mu = 0.043 \text{ mm yr}^{-1} \text{ }^\circ\text{C}^{-2})$	PC prior; $P(a > 0.10) = 10\%$
	b (α_0)	$\text{HN}(\sigma = 0.087 \text{ mm yr}^{-1} \text{ }^\circ\text{C}^{-1})$	Weakly informative; encompasses literature range
	c (trend)	$N(0.3, 1.0^2) \text{ mm yr}^{-1}$	Weakly informative; centered on observed rate
	σ_{extra}	$\text{HC}(\gamma = 5 \text{ mm})$	Model inadequacy
	H_0 (baseline offset)	$N(H_{\text{obs}}(t_1), 5^2) \text{ mm}$	Centered on first observation
<i>Greenland SMB (literature sensitivities, not MCMC)</i>			
	C_T (linear)	$N(-300, 80^2) \text{ Gt yr}^{-1} \text{ }^\circ\text{C}^{-1}$	Inter-RCM spread (MAR, RACMO, HIRHAM)
	C_{T^2} (quadratic)	$N(-50, 30^2) \text{ Gt yr}^{-1} \text{ }^\circ\text{C}^{-2}$	RCM nonlinearity estimates
<i>Greenland discharge (WLS, no explicit priors)</i>			
	γ, r_0	WLS point estimates	Implicit uniform prior via least squares
	δ (delay)	BIC-weighted mixture	$\delta \in \{4, 5, 6, 7, 8\} \text{ yr}$; data-driven
<i>WAIS (scenario mixture, 2 scenarios)</i>			
	$P(\text{S1})$	0.10	Assessed: continued present-day rates
	$P(\text{S2})$	0.90	Assessed: MISI-driven acceleration, with or without MICI
	S1 endpoint	SN _{log} , 25–85 mm	IMBIE extrapolation
	S2 endpoint	SN _{log} ($\alpha = 3.22$), 150–1300 mm	Goldberg et al. lower bound; AR6 low-confidence AIS p95 upper bound
	S2 trajectory β	LN($\ln 1.84, 0.3^2$)	MISI acceleration, probability-weighted blend; Schoof (2007) scaling
	n_{draw} (Mode B)	$N(4.1, 0.4^2)$, $n \geq 3$	Millstein et al.; Ranganathan & Minchew
	r_0 (rheology sensitivity)	0.28	Martin et al.; $R(n) = 1 + r_0(n - 3)$

1.6 Component-level skill metrics

Table S3 summarizes the goodness-of-fit for each independently calibrated component. In addition to these within-sample metrics, Greenland discharge is validated out of sample against the Mankoff et al. (2021) record (1986–2023) and NOAA Arctic Report Card estimates (*Poinar et al.*, 2026), both withheld from calibration; all withheld observations fall within the 90% credible interval (Fig. 2b,c of the main text). Budget closure, the out-of-sample comparison of the summed components against satellite altimetry, is discussed in the main text (Fig. 2f) and provides further confidence that the framework captures the principal contributors to observed sea-level change over the satellite era.

Table S3: Goodness-of-fit for individually calibrated component models. R^2 is computed against the calibration data. The thermosteric model is fitted jointly to two depth ranges. Greenland discharge R^2 is reported in rate space to avoid inflation from cumulative integration (see main text). EAIS and Greenland SMB use literature-derived sensitivities rather than data calibration and are not listed.

Component	Calibration data	R^2	Notes
Thermosteric (0–700 m)	NOAA, 1955–2025	0.965	Two-layer ODE, 6 parameters
Thermosteric (0–2000 m)	NOAA, 2005–2025	0.979	Joint fit with 0–700 m
Glaciers	GlaMBIE, 2000–2023	0.995	Linear rate–temperature
Greenland discharge	Mouginot, 1972–2018	0.794	Rate-space; 3 parameters
Antarctic Peninsula	IMBIE, 1992–2020	0.940	Linear rate–temperature

2 Exceedance Probability by Threshold and Scenario

Table S4 reports the probability of exceeding selected global mean sea-level thresholds (relative to pre-industrial) by 2100 under three warming trajectories, for the full component forecast (which includes the WAIS scenario mixture) and the slow-WAIS scenario alone. Together these columns isolate the contribution of WAIS tail risk: any probability present in the component forecast but absent in the slow-WAIS column is attributable to the possibility of rapid WAIS retreat.

3 Value of Information Analysis

The value of information (VoI) analysis quantifies the economic benefit of resolving the binary uncertainty about whether WAIS retreat remains slow (Scenario 1) or allows for the possibility of rapid retreat (Scenario 2). The framework follows the expected value of sample information (EVSI) formulation from classical decision theory (*Raiffa and Schlaifer*, 1961).

Table S4: **Probability (%) of exceeding global mean sea-level thresholds by 2100.** Thresholds are relative to the pre-industrial baseline (≈ 1850 – 1900). The component forecast column reflects the full WAIS scenario mixture; the slow-WAIS column conditions on the stable-retreat scenario only. Values are computed from posterior samples under each warming trajectory.

SLR threshold (m, pre-ind.)	Warming trajectory	Component forecast	Slow WAIS
0.5	2°C	100.0	100.0
	3°C	100.0	100.0
	4°C	100.0	100.0
0.75	2°C	88.1	0.3
	3°C	99.7	93.5
	4°C	100.0	100.0
1.0	2°C	47.4	0.0
	3°C	75.6	0.1
	4°C	95.5	57.9
1.5	2°C	17.8	0.0
	3°C	23.8	0.0
	4°C	34.9	0.0
2.0	2°C	6.4	0.0
	3°C	9.4	0.0
	4°C	13.9	0.0

3.1 Decision framework

We consider an illustrative coastal adaptation decision where a planner must choose a defense height d (meters of sea-level protection) to minimize total expected costs. The defense is expected to take 20 years to plan and complete construction and to have a minimum 50-year design lifetime ^{***find references***}, giving a target year for planning of 2100. For a given sea-level rise distribution S at 2100, the total annual cost of defense height d is

$$C(d) = C_{\text{adapt}}(d) + \mathbb{E}[D(\max(S - d, 0))], \quad (4)$$

where $C_{\text{adapt}}(d) = c_a \cdot d$ is the annualized adaptation cost and $\mathbb{E}[D(\cdot)]$ is the expected value of the residual damage D from SLR exceeding the defense height (overtopping).

3.1.1 Adaptation cost

The total capital cost of coastal adaptation is estimated at $C_{\text{cap}} = \$4 \text{ T/m}$ of GMSLR. This is derived from Tiggeloven et al. (2020), who estimate that 3.4% of the global coastline ($\sim 54,400 \text{ km}$ of ~ 1.6 million km total) warrants dike heightening under cost-benefit optimality, at a unit cost of $\$7 \text{ M}$ per km per meter of dike height (the median cost raising dikes in New Orleans post-Hurricane Katrina). This yields $\sim \$400 \text{ B/m}$ for dike raising alone along the optimal 3.4% of coastline. We apply a conservative $10\times$ multiplier to account

for the full societal adaptation burden—including seawalls, nature-based solutions, managed retreat, and infrastructure relocation—across a broader fraction of the global coastline.

Because the cost function $C(d)$ (Eq. 4) combines adaptation costs with annual flood damages from Jevrejeva et al. (2018), both terms need to be expressed in consistent units. Adaptation is dominated by capital investment deployed over the planning horizon, so we conservatively annualize C_{cap} using the capital recovery factor (CRF):

$$c_a = C_{\text{cap}} \times \text{CRF}, \quad \text{CRF} = \frac{r(1+r)^n}{(1+r)^n - 1}, \quad (5)$$

where $r = 0.03$ is the real discount rate and $n = 75$ yr is the planning horizon. This gives $\text{CRF} \approx 0.034$ and an annualized adaptation cost of $c_a \approx \$135 \text{ B}/(\text{m}\cdot\text{yr})$ of GMSLR. The resulting $C(d)$ is in $\$/\text{yr}$, the EVSI is in $\$/\text{yr}$, and the NPV conversion (Eq. 9) produces a present-value total in $\$$.

3.1.2 Damage function

Residual flood damages are computed using the piecewise-linear global damage function of Jevrejeva et al. (2018), which maps SLR to annual global flood costs based on estimated asset exposure. For SLR beyond the tabulated range (1.80 m), we extrapolate linearly using the slope of the final segment ($\sim \$13.8 \text{ T}/\text{yr}$ per meter).

3.2 Two-world structure

The forecast distribution is decomposed into two conditional distributions, conditioned directly on A4 scenario membership rather than on a percentile-threshold proxy:

- **Slow WAIS** (S_{slow}) samples from the blended forecast conditional on WAIS Scenario 1 (`S1_status_quo`) representing future GMSLR.
- **Fast WAIS** (S_{fast}) samples from the full forecast conditional on WAIS Scenario 2 (`S2_fast_wais`), representing outcomes that allow for rapid WAIS retreat via MISI, with or without MICI.

Because the A4 framework now comprises exactly these two scenarios, S_{slow} and S_{fast} partition the full mixture completely: every Monte Carlo sample’s WAIS scenario assignment is recorded at the point of sampling and carried through the component-summation and rate-space-blending pipeline as a per-sample label, so the two-world split is exact rather than approximate.

The current prior probability of stable WAIS is $p_s = 0.10$, consistent with our scenario weights (Scenario 1 receives 10% probability).

3.3 EVSI calculation

For a given prior $p_s = P(\text{slow WAIS})$, the EVSI quantifies the annual benefit of learning whether WAIS will lose mass slowly or rapidly before choosing the defense height for coastal development.

3.3.1 Hedging cost (no information)

Without additional information, the planner hedges by choosing the single defense height d_{hedge}^* that minimizes the prior-weighted expected cost:

$$C_{\text{hedge}}^* = \min_d [p_s \cdot C_{\text{slow}}(d) + (1 - p_s) \cdot C_{\text{fast}}(d)], \quad (6)$$

where $C_{\text{slow}}(d) = C_{\text{adapt}}(d) + \mathbb{E}[D(\max(S_{\text{slow}} - d, 0))]$ and $C_{\text{fast}}(d)$ is defined analogously using S_{fast} .

3.3.2 Perfect information cost

With perfect information, the planner would tailor the defense height to whichever world is realized:

$$C_{\text{perfect}}^* = p_s \cdot \min_d C_{\text{slow}}(d) + (1 - p_s) \cdot \min_d C_{\text{fast}}(d). \quad (7)$$

3.3.3 Annual EVSI

The annual value of perfect information is the difference between the hedging cost and the perfect-information cost:

$$\text{EVSI}_{\text{ann}} = \max(C_{\text{hedge}}^* - C_{\text{perfect}}^*, 0). \quad (8)$$

This is the annual cost penalty of having to compromise on a single design rather than tailoring to the true state. When $p_s \approx 0$ or $p_s \approx 1$, the hedging design is already close to the tailored design, so EVSI is small. EVSI is maximized near $p_s \approx 0.5$, where the optimal designs for the two worlds diverge most.

3.4 Discounting and the VoI surface

The annual EVSI is converted to a net present value (NPV) by discounting the stream of benefits from the year information arrives (t_a) to the target year ($t_T = 2100$) at a real discount rate $r = 3\%$:

$$\text{VoI}(t_a, p_s) = \text{EVSI}_{\text{ann}}(p_s) \left(\frac{1 - (1 + r)^{-(t_T - t_a)}}{r} \right). \quad (9)$$

The annuity factor $[1 - (1 + r)^{-n}]/r$ captures the number of discounted years over which the improved decision yields benefits.

The VoI surface (Fig. 4c of the main text) is computed over a grid of arrival years ($t_a \in [2026, 2150]$) and prior probabilities ($p_s \in [0.01, 0.99]$). All calculations use the 3°C (SSP2-4.5) warming trajectory. The choice of warming trajectory has qualitatively little impact on the results given all the uncertainties involved. Note that our cost estimates are likely to be underestimates (see §3.1.1), but the takeaway message remains even if the true costs or damages are an order of magnitude lower.

3.5 Key assumptions and limitations

The VoI calculation is intended to be simple analysis to make a simple point, not to stand for detailed financial decisions. As such, we make several simplifying assumptions:

1. **Binary resolution.** Information perfectly resolves the slow/fast dichotomy. Partial information (e.g., updating p_s from 0.10 to 0.30 without full resolution) would yield lower VoI.
2. **Single decision.** The planner makes one irreversible design choice. In practice, adaptive management with staged decisions would reduce the hedging penalty, lowering VoI.
3. **Global aggregation.** Costs are aggregated globally. Local VoI would differ based on coastal exposure and elevation.
4. **Linear adaptation cost.** We assume C_{adapt} scales linearly with defense height. Economies or diseconomies of scale would modify the optimal designs.
5. **Discount rate.** The 3% real discount rate is standard for infrastructure but lower rates (reflecting intergenerational equity) would increase VoI. Higher discount rates, that would lower VoI, are unlikely given the long timeline of risk exposure from sea level rise.

Despite these simplifications, the analysis provides a rough order-of-magnitude estimate of the economic value of resolving WAIS uncertainty, supporting the argument that targeted investment in ice-sheet observations and process understanding has massive expected returns.

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